

Have your healthcare practitioner fill in the Rx template directly from their computer or print and complete by hand.

Fax completed and **signed prescription** with **medical records**  
to (404) 847-5393.

doctor\_contact\_id:  
outgoing\_fax\_number:

# MOTUS NOVA

## REHAB IN MOTION

### FDA DEVICE INDICATION:

The Motus Hand and Motus Foot are FDA Class 1 at-home neuro rehab robots offering active assistance. Motus technology employs high-dose repetitive task practice to induce neuroplasticity, aiding brain injury survivors in improving arm range of motion and strength with the Motus Hand, and enhancing walking speed and endurance with the Motus Foot.

### PATIENT INFORMATION

FIRST NAME  LAST NAME

ADDRESS  CITY

STATE  ZIP  PHONE NUMBER

ICD-10 CODE(S)  BIRTH DATE  /  /

### HEALTH CARE PRACTITIONER

FIRST NAME  LAST NAME

NPI NUMBER  EMAIL

NATIONAL PROVIDER IDENTIFIER 10-DIGIT NUMBER

ADDRESS  CITY

STATE  ZIP  PHONE

### PRESCRIPTION

Motus Nova Rehabilitation System

Length Of Need:  ☐ 99+

DO NOT LEAVE BLANK OR PUT PRN

#### HAND OR FOOT

- ☐ Motus Foot (E0739)  
☐ Motus Hand (E0739)

#### AFFECTED SIDE

- ☐ Right Side Affected  
☐ Left Side Affected

Reason For Prescription: (Why is this medically necessary? For example: "Motus Hand or Motus Foot would benefit patient because it would help with functional improvements like walking, grip/grasp, and other activities of daily living like eating, dressing, bathing, toileting, transferring, etc.")

DATE  /  /

HEALTHCARE PRACTITIONER SIGNATURE

# Medical Necessity and Clinical Efficacy

## Neuro-Rehabilitation Evidenced-Based Practices:

High intensity repetitive task practice delivered via robot-assisted therapy is recommended to improve motor function in individuals in sub-acute and chronic care settings. These therapies have achieved the highest level of evidential support by the American Heart Association (Class I, Level of Evidence A) and the Cochrane Review 1,2 . The Motus Foot provides inclusive rehabilitation option for patients of varying levels of movement following neurologic injuries, even those with very limited movement to acquire high-intensity repetitive task practice. Clinical trials have shown efficacy across different episodes of care. The Motus Foot provides an effective means to augment each patient's rehabilitation using the brain's own inherent neuroplasticity.

## Number of Studies:

To date, the Motus Hand and Motus Foot® have been variously utilized in 24 peer-reviewed publications ranging from engineering sensor design, basic cortical network plasticity, multi-site randomized control trials, to large multi-state healthcare system implementation studies.

## Chief Findings:

Arm and Hand Function: Motus Hand improves active and passive range of motion, fine and gross motor function, and strength 3-8 .

## Activities of Daily Living:

Following 3-months of Motus Hand rehabilitation, stroke survivors increased their levels of independence in daily activities (required less assistance from either a person or device to accomplish daily tasks) 7,8. Lower Extremity Function: Following 3-months of Motus Foot rehabilitation, stroke survivors have increased dorsiflexion force that is maintained over 4 weeks following the intervention. The Motus Foot® improves gait speed and walking endurance (distance) in chronic stroke survivors 4,9,10.

## Cortical Networks:

Positive changes in cortical network behavior (as assessed by dynamic causal modeling) indicate that 3-months of Motus Hand rehabilitation can assist with normalization of cortical function 11.

## Quality of Life:

The Motus Hand and Motus Foot improve quality of life and reduces depression symptoms 4,5,7,12.

## Cost:

The Motus Hand and Motus Foot have been shown to reduce cost of delivering stroke rehabilitation by between 55-65% to the healthcare system. Savings were realized through large reduction in transportation and time-costs of delivering care in a traditional clinical setting 4,7.

## Combination Therapies:

The Motus Hand works synergistically with existing neurorehabilitation options in two primary ways. Data support ability to amplify functional improvements when combined with standard clinical care. The Motus Hand can also act as a bridge therapy (alternative) when insurance benefits are exhausted 3,5,6.

## Efficacy in Home:

To date, the Motus Hand and Motus Foot are the only robotic based neurorehabilitation therapies that have been fully deployed in the home environment. Further, their therapeutic index and safety profiles remain high even when utilized without real-time oversight of a licensed therapist 4-7,12.

## References

- 1 Winstein, C. J. et al. Guidelines for adult stroke rehabilitation and recovery: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke* 47, e98-e169 (2016).
- 2 Mehrholz, J., Pohl, M., Platz, T., Kugler, J. & Elsner, B. Electromechanical and robot-assisted arm training for improving activities of daily living, arm function, and arm muscle strength after stroke. *Cochrane Database of Systematic Reviews* (2018).
- 3 Kutner, N. G., Zhang, R., Butler, A. J., Wolf, S. L. & Alberts, J. L. Quality-of-life change associated with robotic-assisted therapy to improve hand motor function in patients with subacute stroke: a randomized clinical trial. *Physical therapy* 90, 493-504 (2010).
- 4 Housley, S. et al. Increasing access to cost effective home-based rehabilitation for rural veteran stroke survivors. *Austin journal of cerebrovascular disease & stroke* 3, 1 (2016).
- 5 Linder, S. M. et al. Improving quality of life and depression after stroke through telerehabilitation. *American Journal of Occupational Therapy* 69, 6902290020p6902290021-6902290020p6902290010 (2015).
- 6 Wolf, S. L. et al. The HAAP (Home Arm Assistance Progression Initiative) trial: a novel robotics delivery approach in stroke rehabilitation. *Neurorehabilitation and neural repair* 29, 958-968 (2015).
- 7 Butler, A. J. et al. Expanding tele-rehabilitation of stroke through in-home robot-assisted therapy. *Int J Phys Med Rehabil* 2, 1-11 (2014).
- 8 Rosenstein, L., Ridgel, A. L., Thota, A., Samame, B. & Alberts, J. L. Effects of combined robotic therapy and repetitive-task practice on upper-extremity function in a patient with chronic stroke. *American Journal of Occupational Therapy* 62, 28-35 (2008).
- 9 J. Lynskey, B. T., G. Abruzzo. Home-based robot-assisted ankle rehabilitation for chronic stroke survivors. (2014).
- 10 Peterson, S. Home-based Robot Assisted Ankle Rehabilitation for Chronic Stroke Survivors. (2014).
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- 12 Cherry, C. O. B. et al. Expanding stroke telerehabilitation services to rural veterans: a qualitative study on patient experiences using the robotic stroke therapy delivery and monitoring system program. *Disability and Rehabilitation: Assistive Technology* 12, 21-27 (2017).